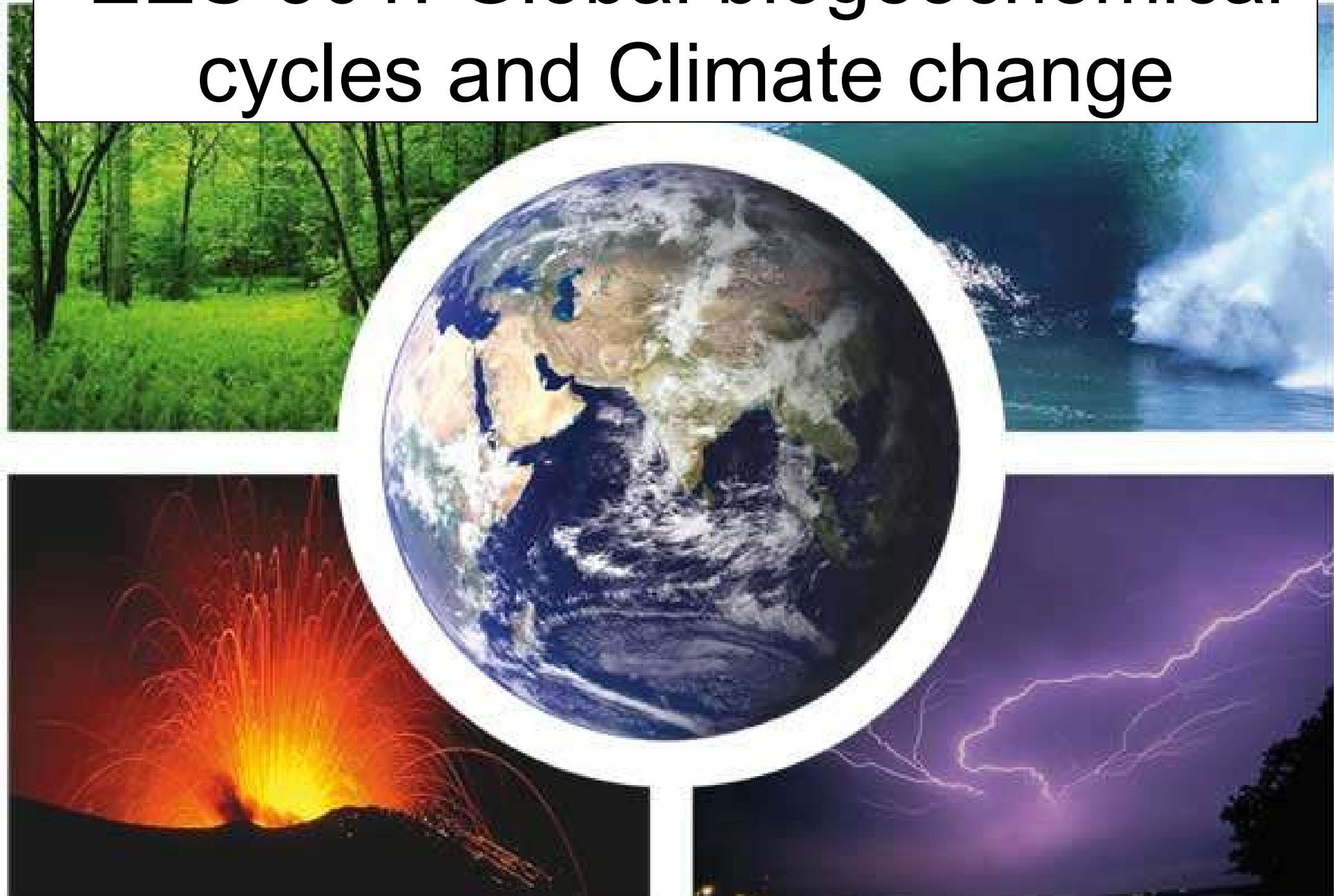


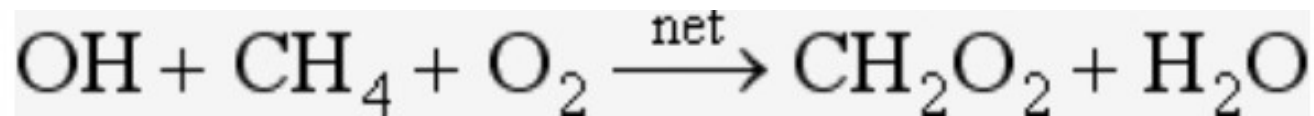
# EES 301: Global biogeochemical cycles and Climate change



# Chemical feedbacks in the climate system

Chemical feedbacks can amplify or slow climate change depending on their natures.

➤ Methane which is primarily lost via reaction with the OH radical an increase in the atmospheric burden results in a higher OH loss rate and lower OH concentration which increases the methane lifetime.



➤ Methane is also the primary source of stratospheric water vapor which is a GHG. Hence increase in methane emissions increases water vapor forcing.

➤ Methane emissions can impact polar stratospheric cloud formation via their impact on stratospheric water vapor and can negatively impact the stratospheric ozone layer. This may reduce the GHG effect due to O<sub>3</sub> at the expense of life on land.

➤ CO<sub>2</sub> solubility in ocean water is temperature and pH dependent. Higher temperatures will lower CO<sub>2</sub> solubility increasing the fraction retained in the atmosphere and hence the GHG effect.

➤ Higher CO<sub>2</sub> emissions and ocean uptake will result in ocean acidification which will lower CO<sub>2</sub> solubility increasing the fraction retained in the atmosphere and hence the GHG effect.

# Biochemical feedbacks in the climate system

Warming wetlands have higher methane emission rates. This is a biogeochemical feedback which increases warming further.

Melting permafrost is typically converted into wetlands, which emit methane and carbon dioxide. This is a biogeochemical feedback which increases warming further.

Algae blooms can increase ocean albedo.

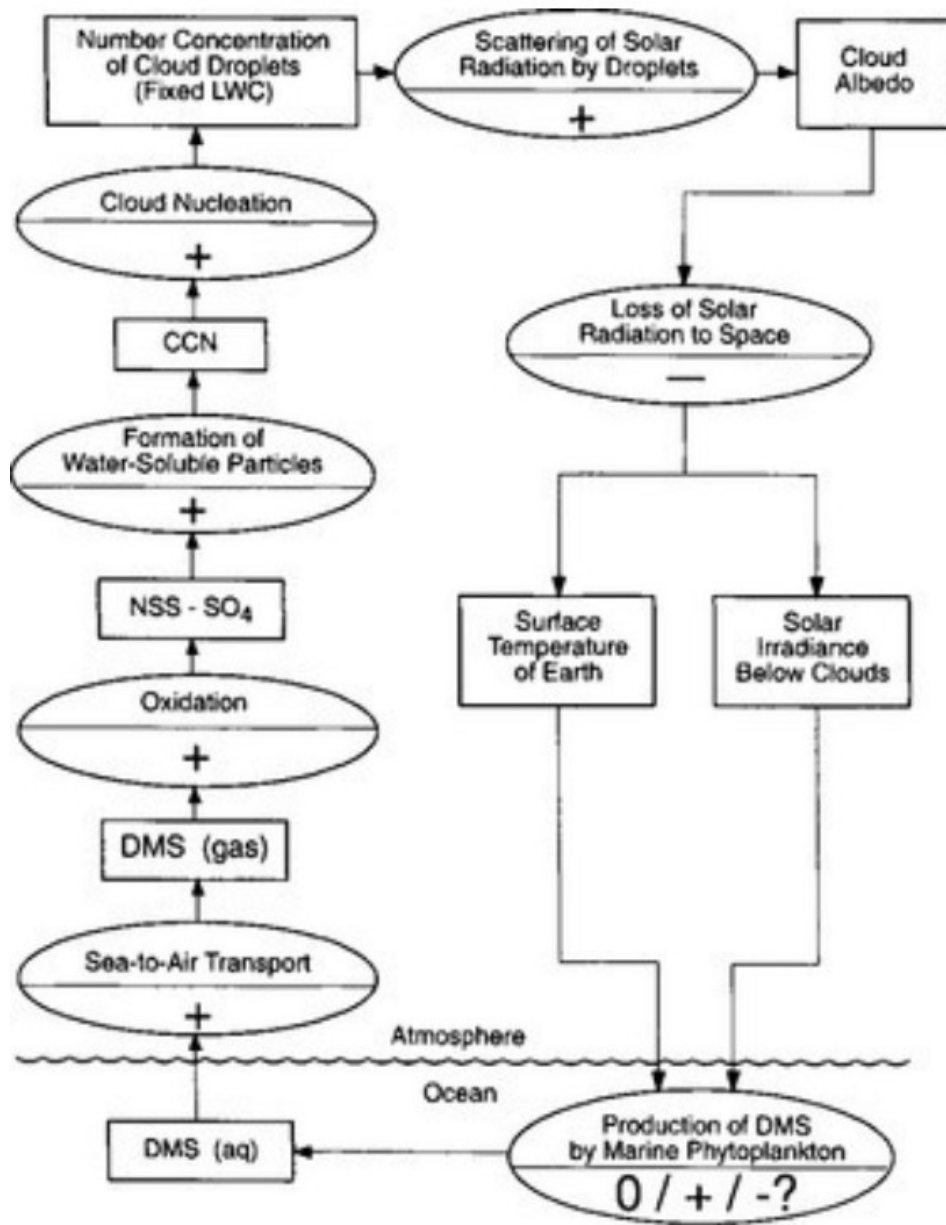
DMS emissions from plankton increase cloud cover over the ocean

CO<sub>2</sub> fertilization can marginally increase terrestrial primary productivity

Evapotranspiration and biogenic OA emissions as well as SOA formation from biogenic VOC emissions may increase cloud cover over forests increasing albedo and cooling the surface.

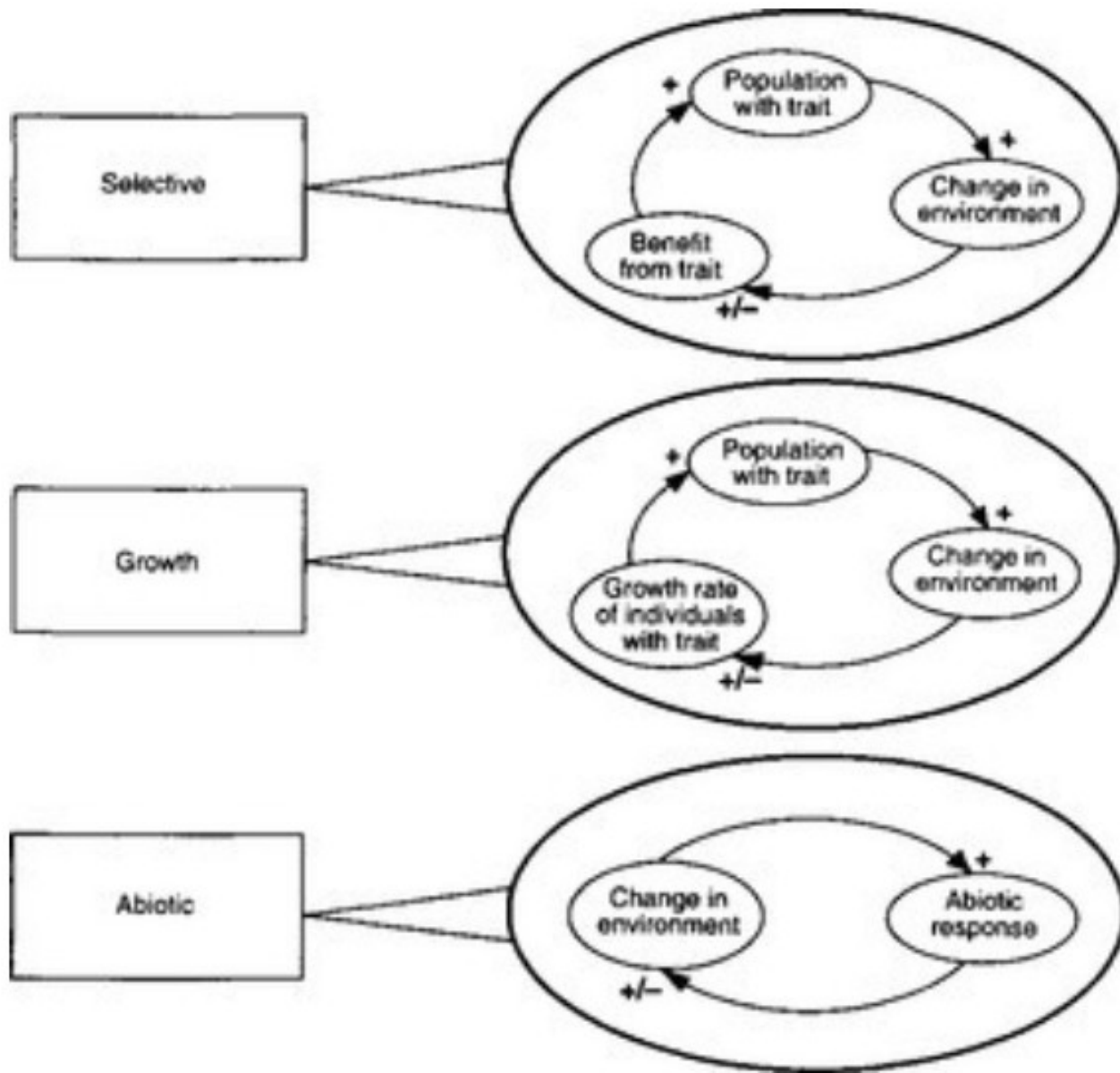
Fungal spores are efficient IN and can trigger rainfall

# Biochemical feedbacks in the climate system



**FIG. 17-9** A hypothetical feedback loop indicating dimethylsulfide (DMS) from marine phytoplankton. The rectangles are measurable quantities, and the ovals are processes linking the rectangles. The sign (+ or -) in the oval indicates the effect of a positive change of the quantity in the preceding rectangle on that in the succeeding rectangle. The most uncertain link in the loop is the effect of cloud albedo on DMS emission; its sign would have to be positive in order to regulate the climate. (From [Charlson et al. \(1987\)](#) . Reprinted by permission from Nature, Copyright © (1987) Macmillan Magazines Ltd.)

# Gaya hypothesis biogeochemical feedbacks influence evolution



There is some evidence in favor of the selective model in particular when it comes to land biota with the ability to trigger/sustain rain that evolved in semi-arid climate zones

**FIG. 17-10** A hierarchy of environmental feedbacks. Three levels are identified, abiotic (purely geochemical and geophysical) feedback, feedback on growth, and feedback on natural selection (see [Chapter 3](#)). “Trait” always refers to an environment-altering trait, and “growth” includes reproduction. At each level, both positive and negative feedbacks are possible, and these are illustrated in



# The geological record of climate change

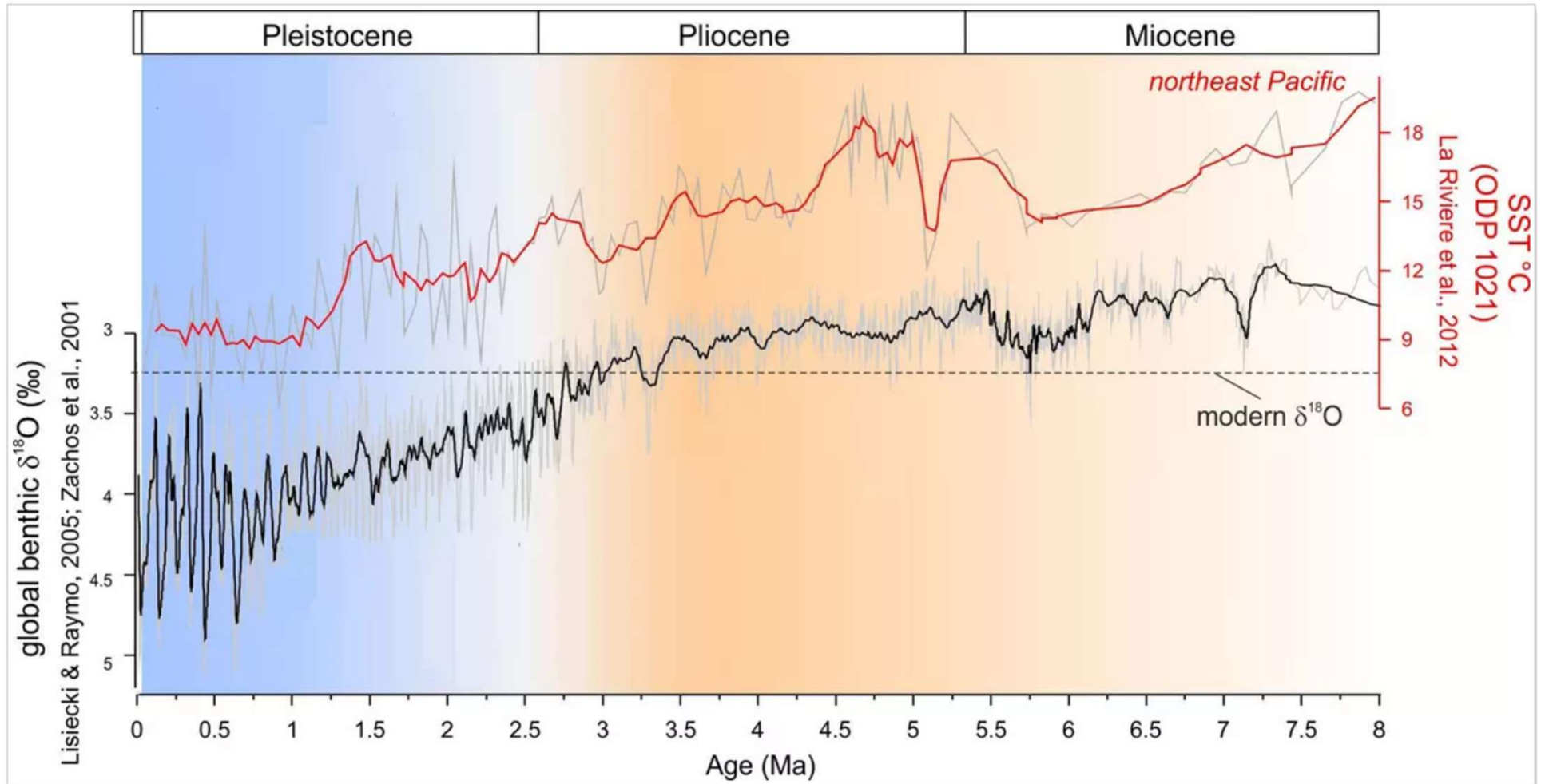


In the mid-19<sup>th</sup> century geologists started to realize that many features of the landscapes in Eurasia and Northern America could be sensibly explained only as the product of glaciations (Agassiz 1840)

These features include bouldery ridges called moraines found far from currently glaciated areas, enormous boulders found far from their source, deep U shaped lake basins, very large ancient river channels and aeolian deposits (loess plateaus).

The revelation that densely populated regions of Europe and North America were rather recently buried by ice led to the startling realization that Earth can experience environmental changes of spectacular magnitude and global significance. Careful analysis of deposits revealed several major named glaciation events.

# The geochemical record of climate change



Starting from the 1950s new geochemical and paleontological tools started to paint a clearer picture of these ice ages. In particular the oxygen isotopic composition of biota trapped in ocean-floor deposits is a measure of continental ice-sheet volume. This data revealed ~30 major glaciation events over the past 2 million years that occurred with startling regularity.

# Proxies and archives

Past variations in climate are inferred from proxy records. A proxy is a physical or geochemical parameter that was in some way influenced by the environmental condition at the time of its deposition. Proxy include the oxygen isotopic composition of sediment deposits in the ocean, the oxygen and hydrogen isotopic composition of the ice in ice cores, the abundance and species composition of plant and animal fossils trapped in sediments, the width of tree rings in fossil wood, gasses and aerosols trapped in ice cores, morains and aeolian deposits deposited on continents during ice ages, drop stones documenting ice-berg calving deposited in marine sediments...

A proxy is typically recovered from archives. Archives can include ice cores, lake deposits, ocean sediment cores or terrestrial sediment cores (e.g. from peat deposits).



Archive: the  
Antarctic ice core

## Proxies:

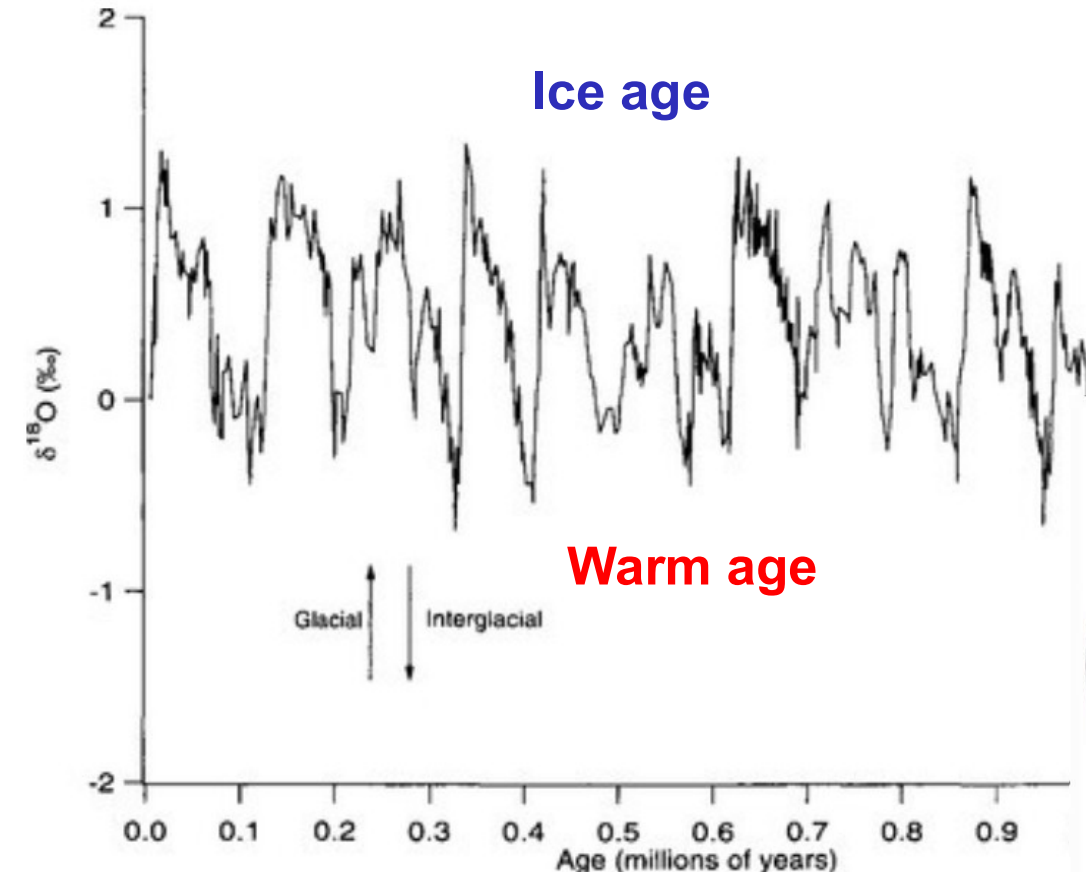
- 1) The trapped dust aerosol
- 2) the trapped gas bubbles which preserve ancient  $\text{CO}_2$  and  $\text{CH}_4$  mixing ratios
- 3) the O and D isotopic composition of the ice ...



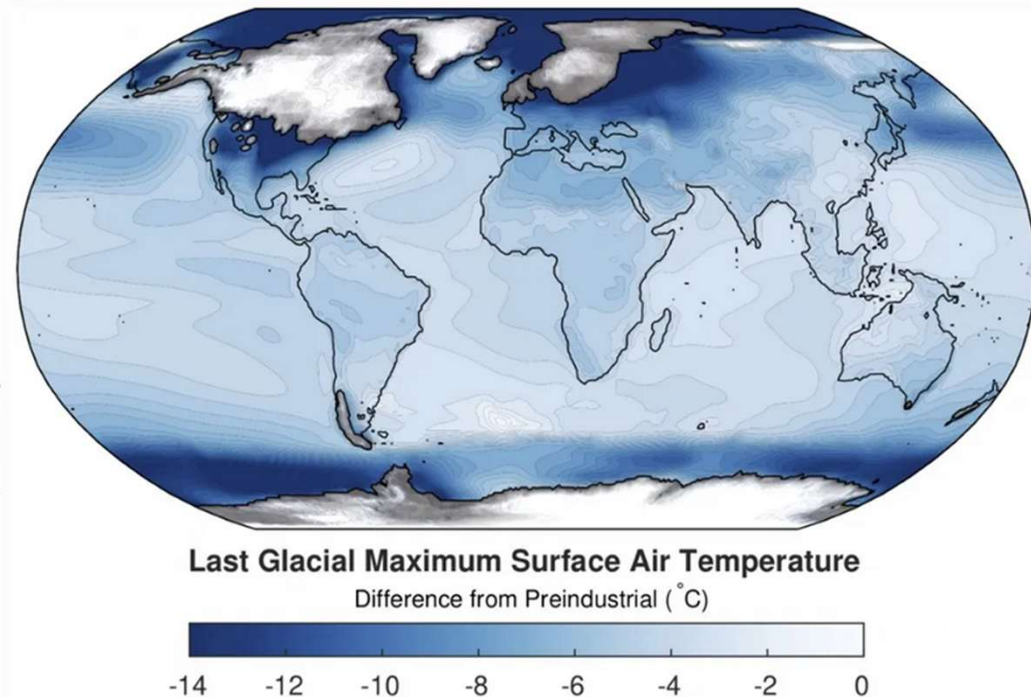
# Quarternary climate change

The main cycle is of ~100,000 year duration, and manifest with a gradual descent into cold climates followed by an abrupt termination of glacial conditions.

In the quaternary the climate alternated between glacials in which much of the northern hemisphere was covered by ice sheets and sea levels were ~200 m lower and interglacials experiencing conditions similar to the pre-industrial Holocene.

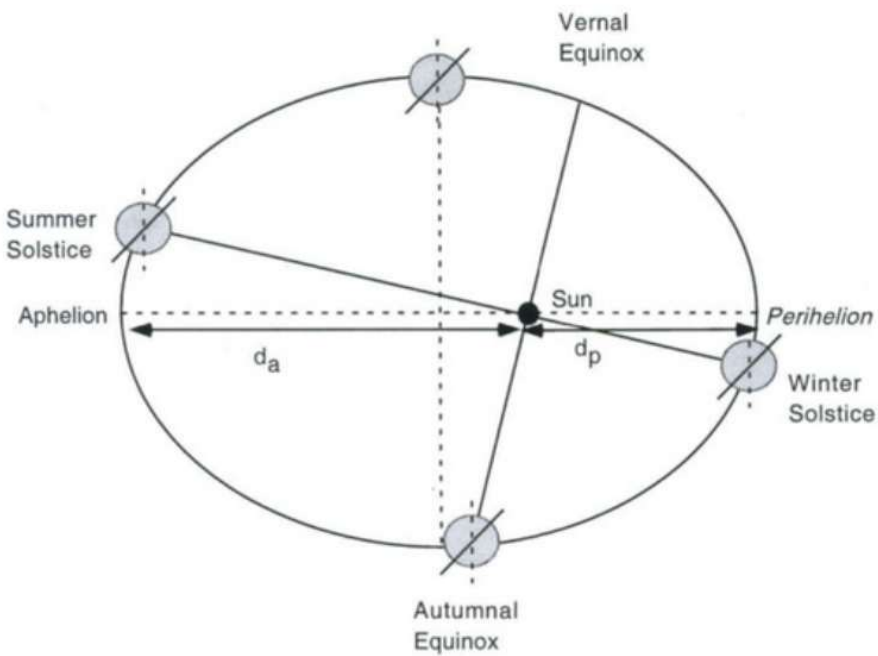


**FIG. 18-1** Benthic foraminiferal oxygen isotope record from 3477 m water depth in the eastern tropical Pacific ocean from Ocean Drilling Program site 677 ([Shackleton et al., 1990](#)).  $^{18}\text{O}/^{16}\text{O}$  ratios are expressed in the  $\delta$  notation relative



This global map indicates the temperature differences between now and preindustrial times, where dark blue translates to cooler temperatures.  
Jessica Tierney, via University of Arizona

# Milakovich cycles



Milankovitch cycles named after the Serbian geophysicist and astronomer Milutin Milanković describe how variations in eccentricity (which varies over 100,000 yrs from nearly 0.0034 to almost 0.058), axial tilt (which varies between  $22^{\circ}2'33''$  and  $24^{\circ}30'16''$ , with a mean period of 41,040 years), and precession (which goes through one cycle every 26,000 years) combine to result in cyclical variations in the intra-annual and latitudinal distribution of solar radiation at the Earth's surface that influenced the Earth's climatic patterns.

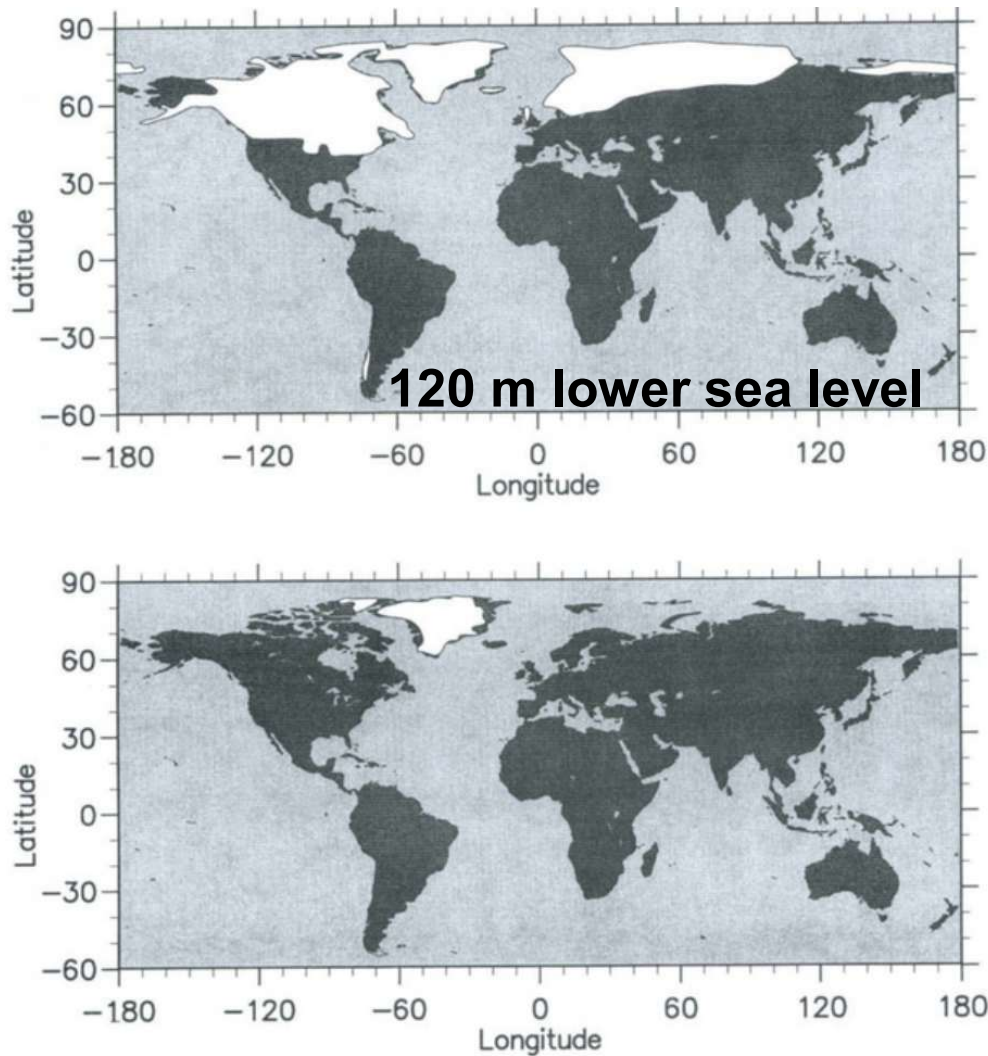
**FIG. 18-2** Orbital parameters that control the seasonal and annual receipt of radiation at the Earth's surface. The Earth's rotational axis tilts and wobbles as shown. The eccentricity of the orbit is  $(d_p - d_a)/(d_p + d_a)$  which is zero for a circle but for the Earth's orbit is sometimes as large as 0.06. (After [Hartmann \(1994\)](#), p. 303) and [Turekian \(1996, p. 80\)](#).)

The precision cycle leads to colder northern hemisphere summers and warmer southern hemisphere summers when the summer solstice occurs near the aphelion. Reverse conditions prevail when the northern hemnspheric summer occurs near the perihelion.

Tilt variation alter the duration of polar summer and winter and lead to warmer summers and colder winters for both hemispheres.

Eccentricity leads to a  $0.5 \text{ W/m}^2$  change in the annual average radiation received with the total amount highest when the orbit is most elliptical (0.058). It also changes the duration of seasons.

# Ice Age Earth



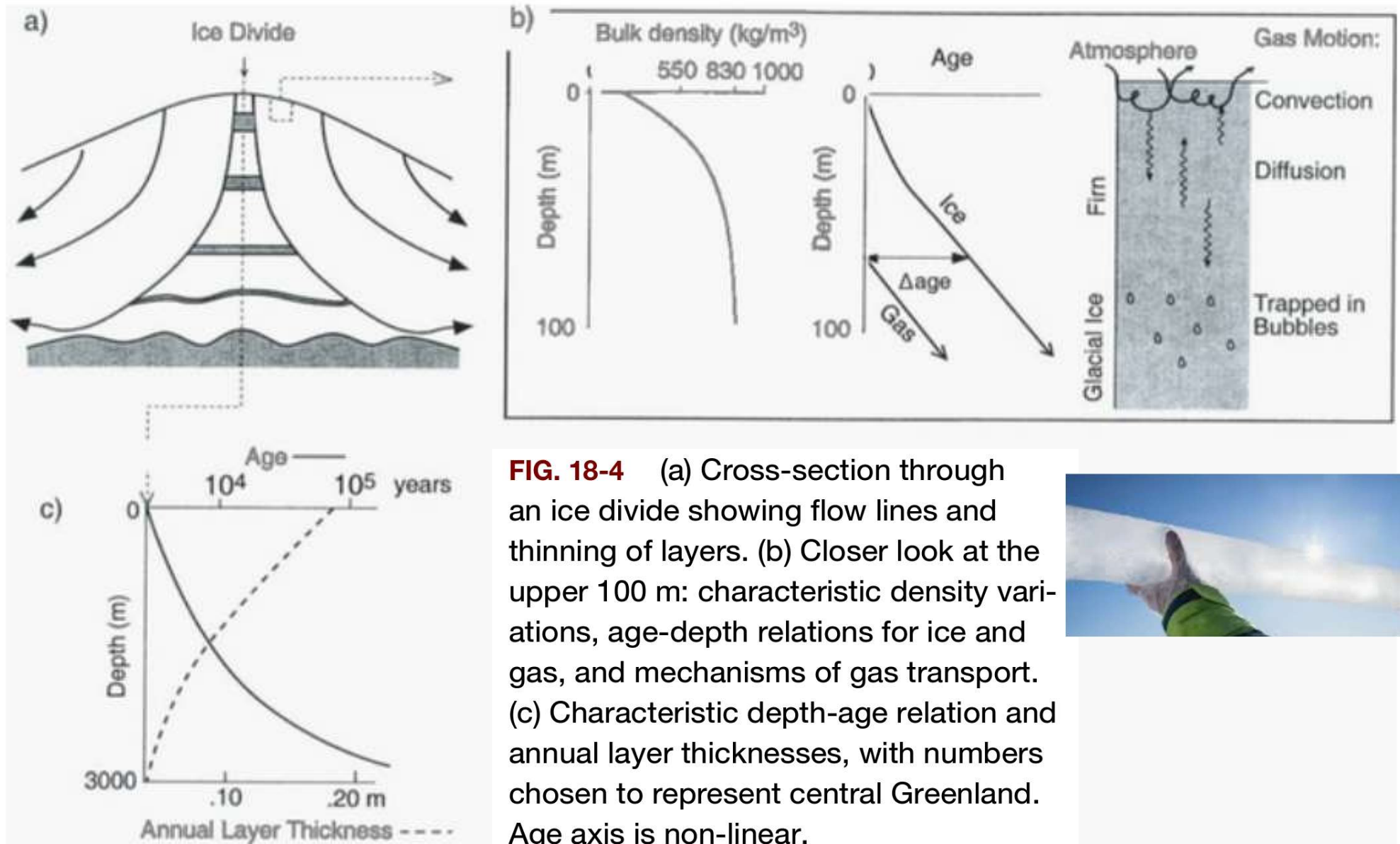
**FIG. 18-3** Extent of ice sheet cover (shown as white) during the LGM (top

- Many areas were dryer and the hydrological cycle was suppressed.
- This is supported by aeolian deposits like sand dunes and loess plateaus deposited across Central and Southern Europe, Russia, Mongolia, China and the Midwest in the US as well as pollen records
- Pollen –based reconstructions of mid-high-latitude land temperatures suggest a temperature depression by 4 to 8°C and an equatorward shift of vegetation zones
- High-latitude seas surface temperatures appear to have been depressed by 6 to 10°C.
- Glacial deposits on tropical mountains indicate snowlines were depressed by 1000 m corresponding to a 5-6 °C temperature depression of tropical land temperatures.
- Proxies for North Atlantic Deep water formation in the form of the Cd/Ca ratio and  $\delta^{13}\text{C}$  of benthic foraminifera indicate that poleward heat transport by the Gulf stream was greatly reduced during glacials



# Ice Core basics

➤ The age of the ice below the divide increases exponentially with time. Each layer becomes thinner the older it gets

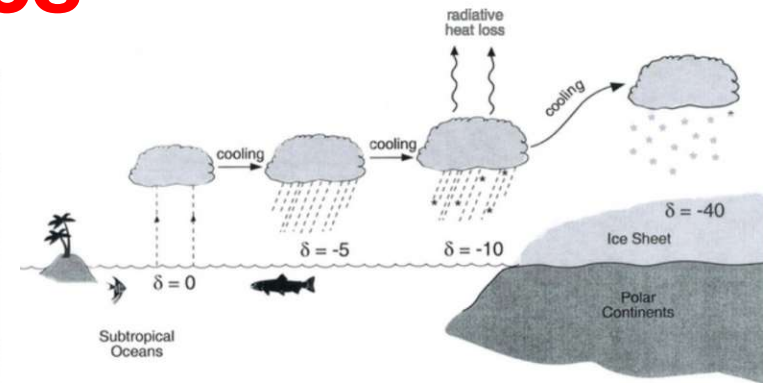
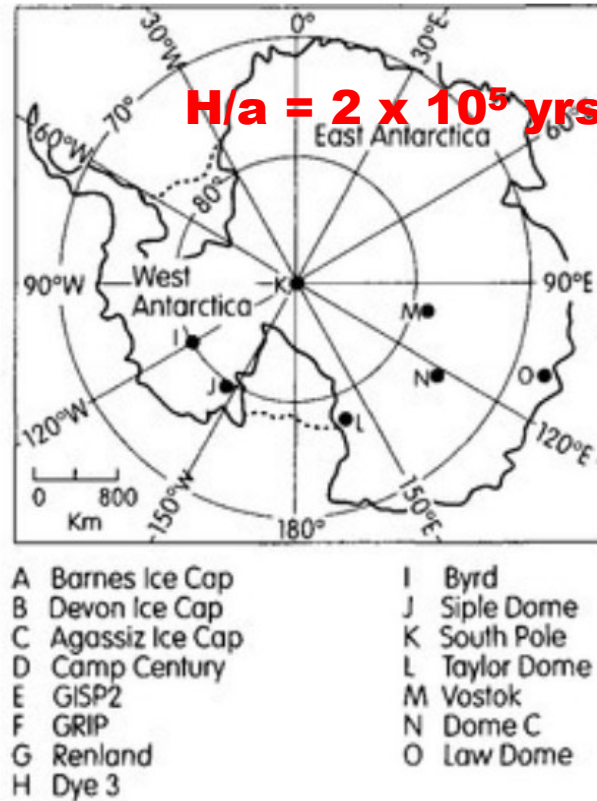
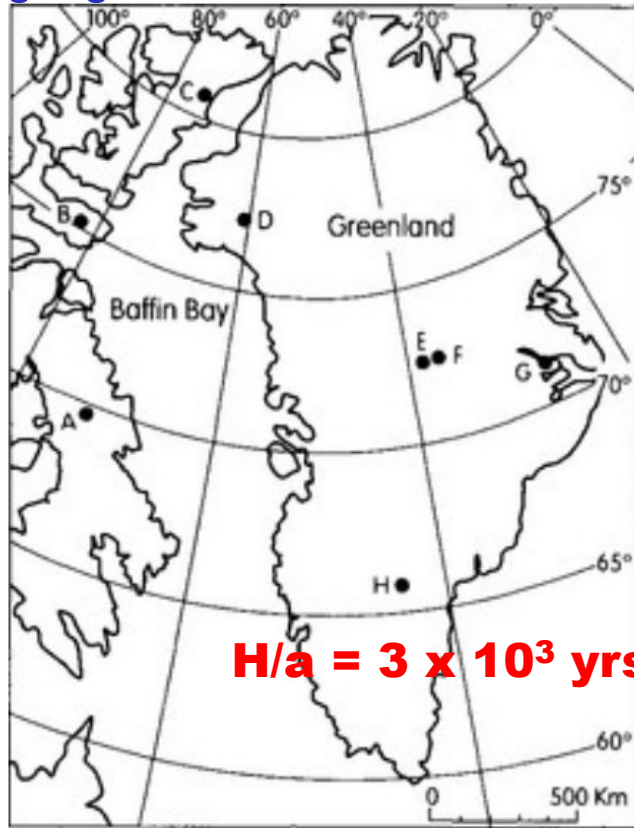


**FIG. 18-4** (a) Cross-section through an ice divide showing flow lines and thinning of layers. (b) Closer look at the upper 100 m: characteristic density variations, age-depth relations for ice and gas, and mechanisms of gas transport. (c) Characteristic depth-age relation and annual layer thicknesses, with numbers chosen to represent central Greenland. Age axis is non-linear.



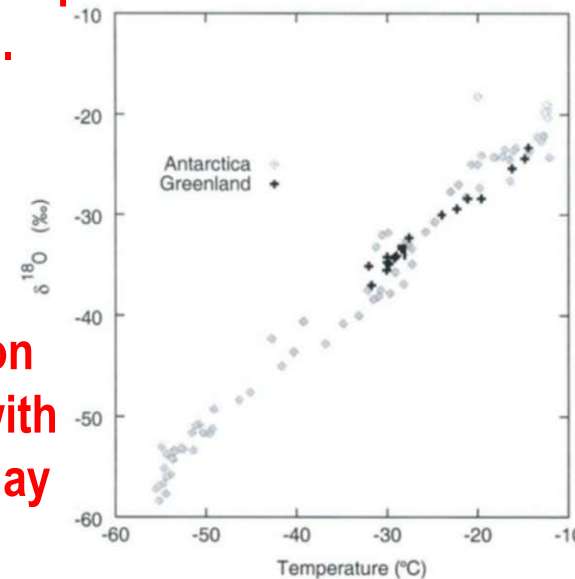
a is an ancient precipitation gauge for the core site.

# Ice core basics



**FIG. 18-6** Characteristic air-mass trajectory and corresponding per mil isotopic composition of precipitation, along a transect from the subtropics to a polar ice sheet. This is a highly schematic view of the true atmospheric system.

**Reconstruction of past temperature is done by measuring the Oxygen and D isotopic composition of the water molecules.**



**Calibration is done with present day snow**

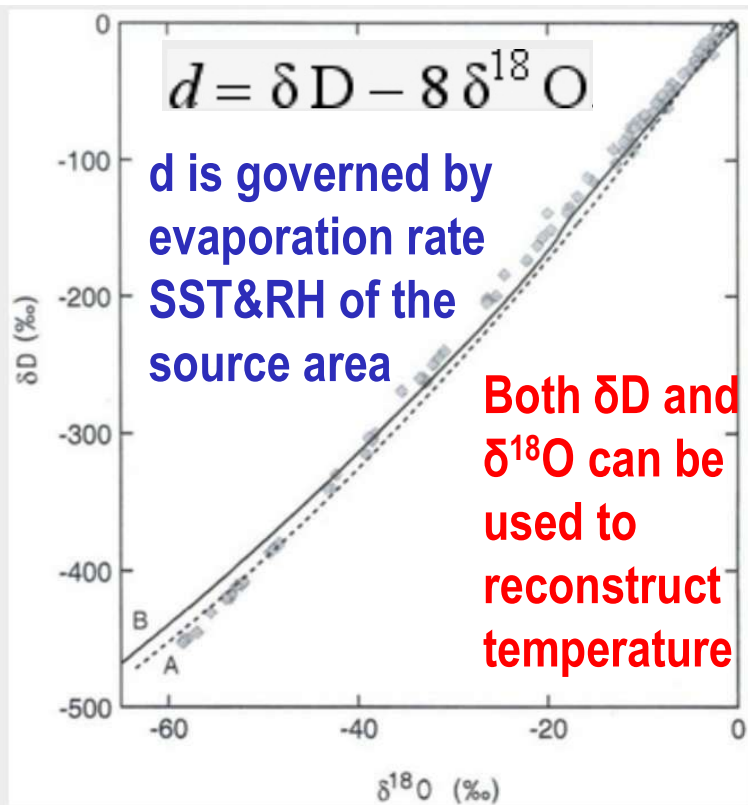
**FIG. 18-5** Locations of important ice-core sites in Greenland and Antarctica. The two central Greenland sites, GISP2 and GRIP, are collectively known as Summit. (after [Paterson, 1994](#) ).

**Dating is possible through annual layer counting, identification of the fallout from major known volcanic eruptions and with absolute dating methods**

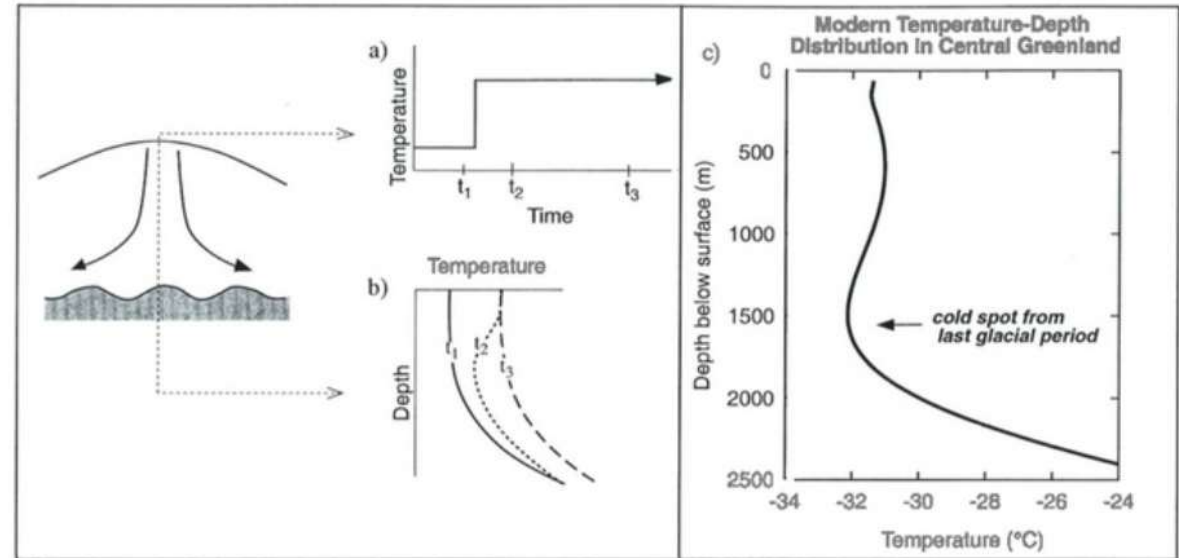
**Ice sheets with a large thickness H in relation to the accumulation rate a (unit of thickness deposited per unit time) have long records but poor temporal resolution**



# Ice core basics



**FIG. 18-24** Observed correlation (the Meteoric Water Line) of the two most important isotopic ratios in precipitation (gray diamonds; [Jouzel et al., 1987](#) and [Dahe et al., 1994](#)), and predictions of simple isotopic models. A, prediction with constant  $\alpha$ ; B,



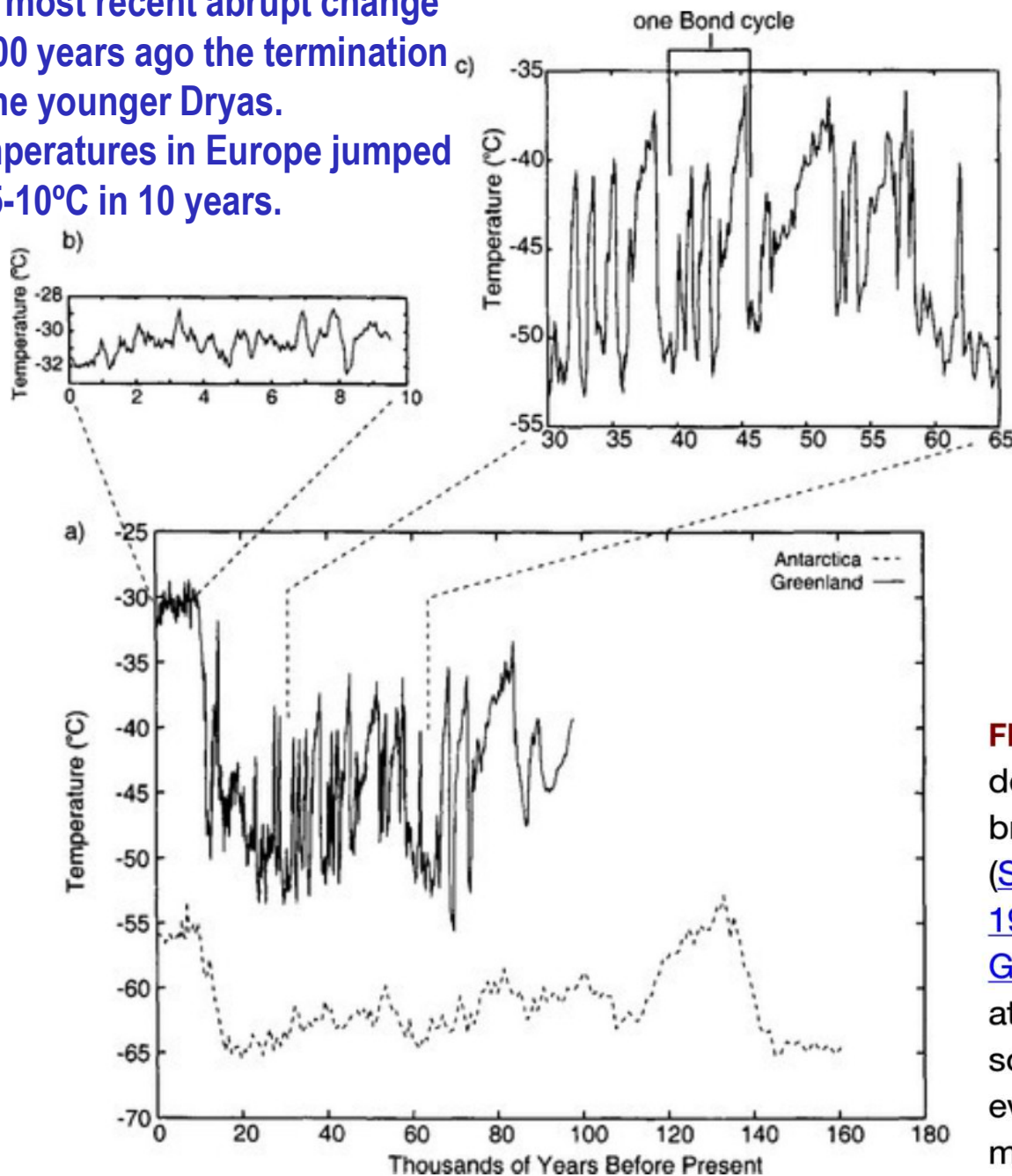
**FIG. 18-8** Characteristic temperature–depth distributions at an ice divide. For a climatic temperature history as shown in (a) the temperature–depth distribution changes as shown in (b). Following the step increase in surface temperature, the initial steady temperature profile ( $t_1$  in (b)) is altered by a warming wave (e.g., at time  $t_2$ ) but eventually reaches a new steady profile by time  $t_3$ . (c) Temperature

Borehole temperature depth structure can be used to reconstruct the long-term average temperatures of the last glacial and can be used to calibrate the stable isotope records for the borehole location

Trapped gases are not of the same age as the ice, as gasses move through firn. Rapid climate change will cause thermal gradients and an isotope fraction of the  $\delta^{15}N$  in  $N_2$  gas that is proportional to the ancient temperature change. The depth separation between the gas-phase  $\delta^{15}N$  spike and the corresponding  $\delta^{18}O$  record allows  $\Delta_{ade}$  calibration

# Temperature records

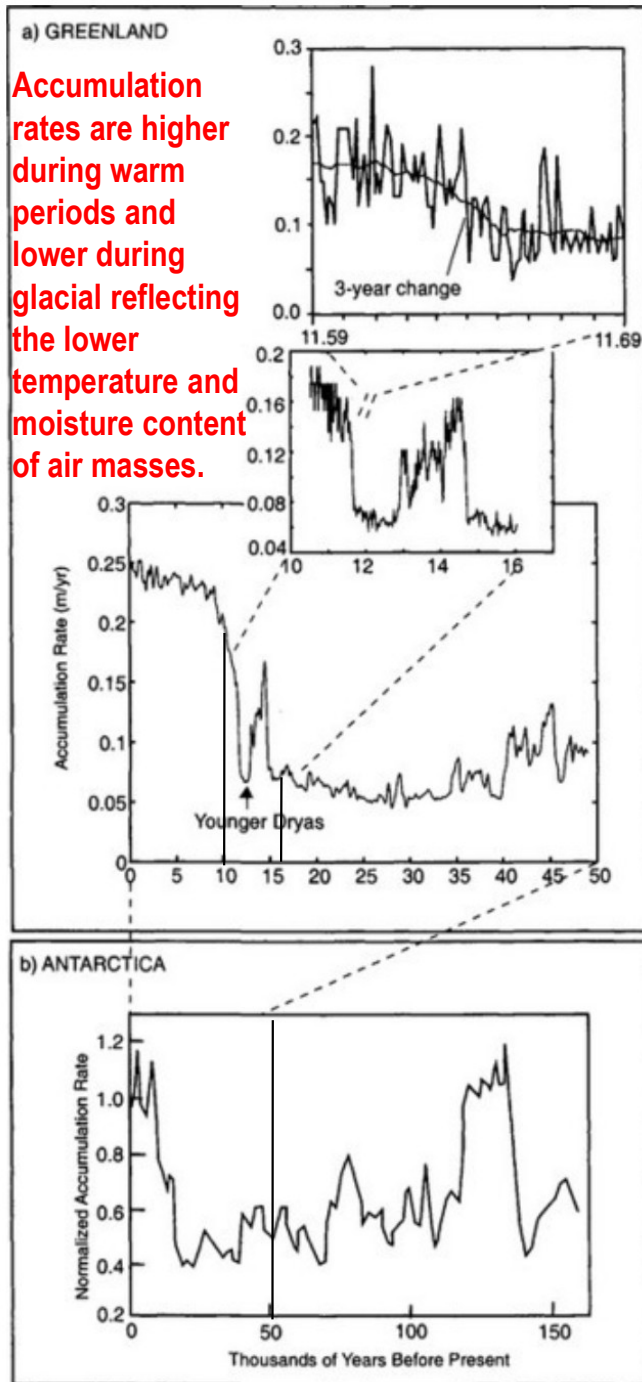
The most recent abrupt change 11500 years ago the termination of the younger Dryas. Temperatures in Europe jumped by 5-10°C in 10 years.



The rapid spikes within glacials are called Dansgaard-Oeschger events and the cold extremes of these events are related to rapid increases in the iceberg discharge from the North American ice sheets into the North Atlantic (Heinrich event) and disruptions in the North Atlantic Deep water formation that slowed the gulf stream. Events are characterized by abrupt warming (restoration of AMOC) followed by gradual cooling (slowing of AMOC).

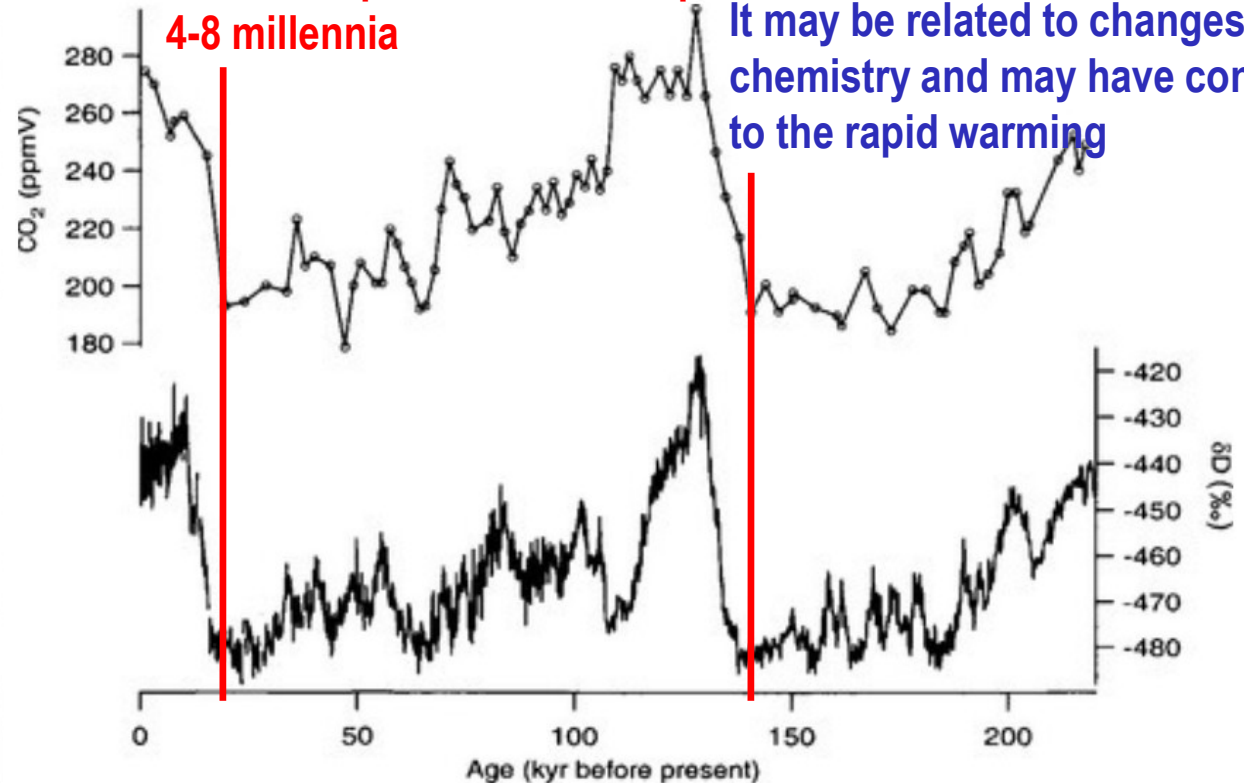
**FIG. 18-9** (a) Temperature histories derived by borehole-temperature calibration of isotopic records, for Antarctica ([Salamatin et al., 1998](#); [Jouzel et al., 1993](#)) and Greenland ([Cuffey et al., 1995](#); [Grootes et al., 1993](#)). (b) A closer look at the Holocene. (c) A closer look at some of the rapid climate changes (D-O events) during the last glacial period. The magnitudes of these temperature swings are not yet calibrated, and may be only half as large as depicted here.

# Accumulation rate records & chemical tracers



Interglacial  $\text{CO}_2$  mixing ratios were 250-280 ppm. Glacial  $\text{CO}_2$  mixing ratios were lower ~180-200 ppm as  $\uparrow$  dust transport iron fertilized oceans and  $\text{CO}_2$  solubility increased at lower SSTs.

The transition to higher  $\text{CO}_2$  mixing ratios at the start of the Holocene preceded the temperature rise at ice core drill sites by 4-8 millennia. It may be related to changes in ocean chemistry and may have contributed to the rapid warming.

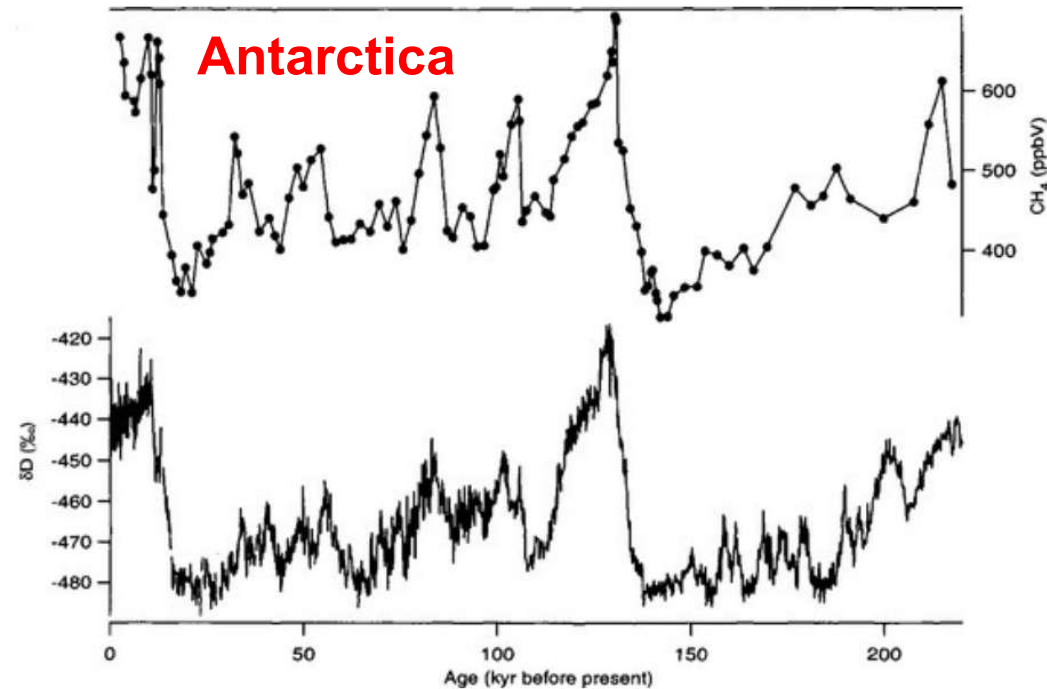


**FIG. 18-10** (a) Accumulation rate history for Greenland ([Alley et al., 1993](#); [Cuffey and Clow, 1997](#)), with insets focusing on the Younger Dryas cold interval and the very rapid termination of the last glaciation, with substantial change that occurred in only 3 years noted, derived

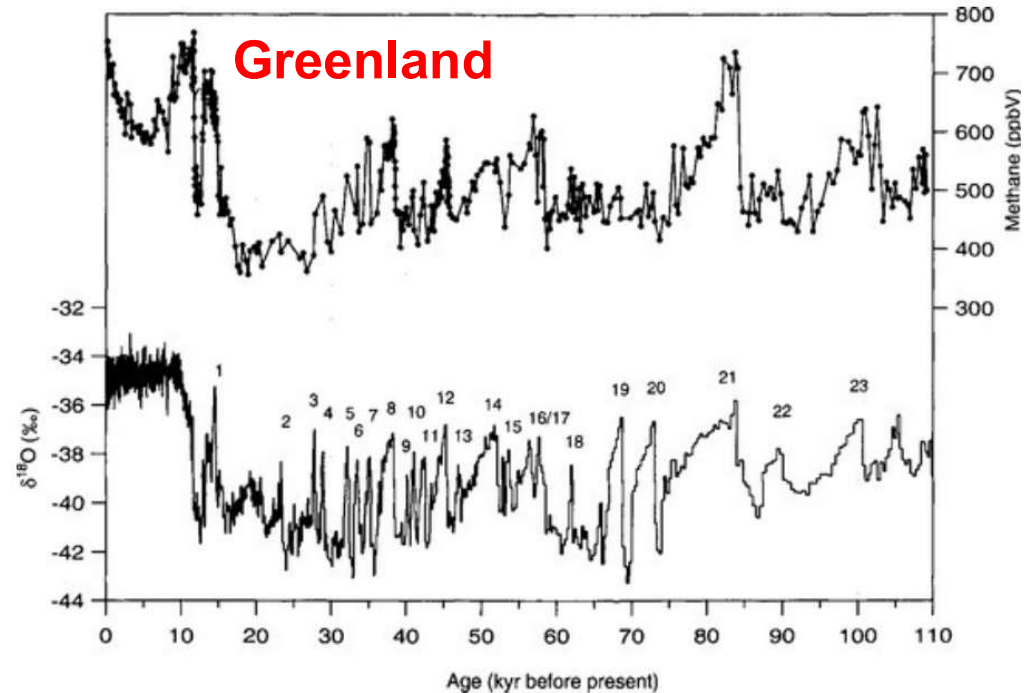
**FIG. 18-12** Record of atmospheric  $\text{CO}_2$  over the past 220 000 years from the Vostok ice core ([Jouzel et al., 1993](#); [Barnola et al., 1987](#)), and the corresponding deuterium isotope record temperature proxy ([Jouzel et al., 1993](#)) (see [Section 18.3.2](#)).



# Ice age atmospheric composition



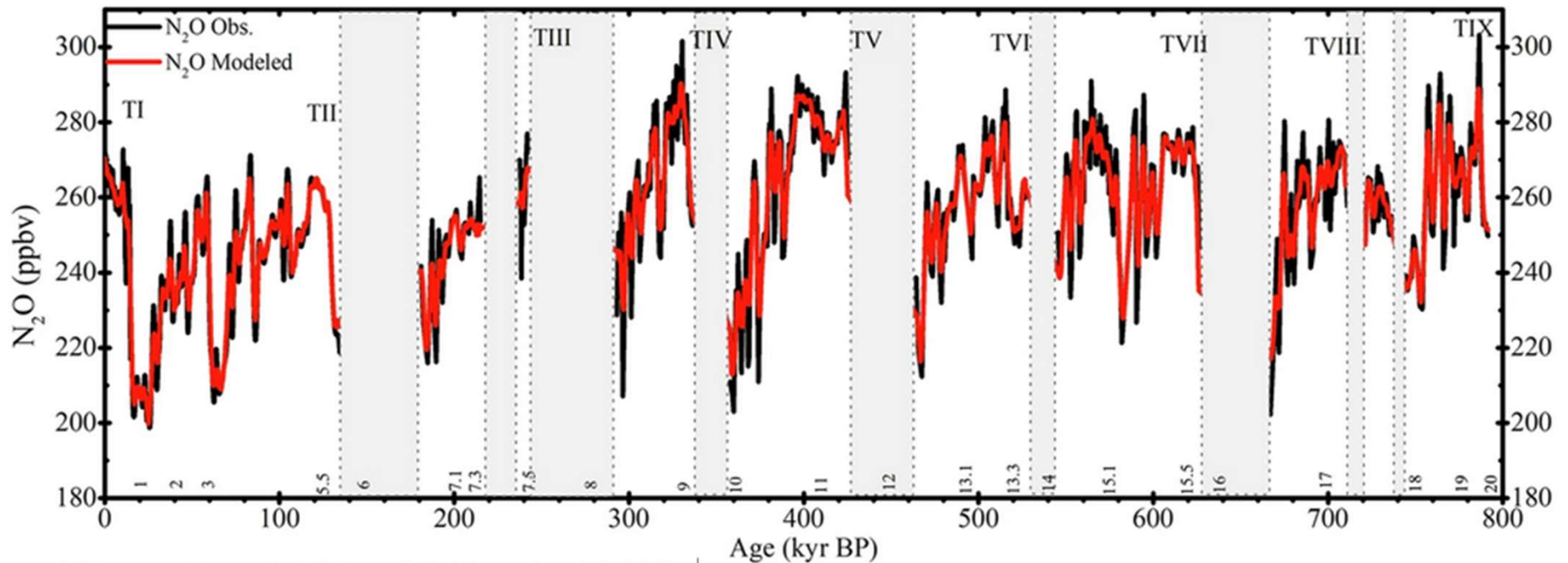
**FIG. 18-15** Ice-core methane record from Vostok, Antarctica, for the past 220 000 years ([Chappellaz et al., 1990](#); [Jouzel et al., 1993](#)).



**FIG. 18-16** Methane record for the past 110 000 years from the GISP2 ice core ([Brook et al., 1996](#); Brook et al., in press). Numbers are labels for each of the warm “interstadial” periods within the cold glacial climate.



# Ice age atmospheric composition

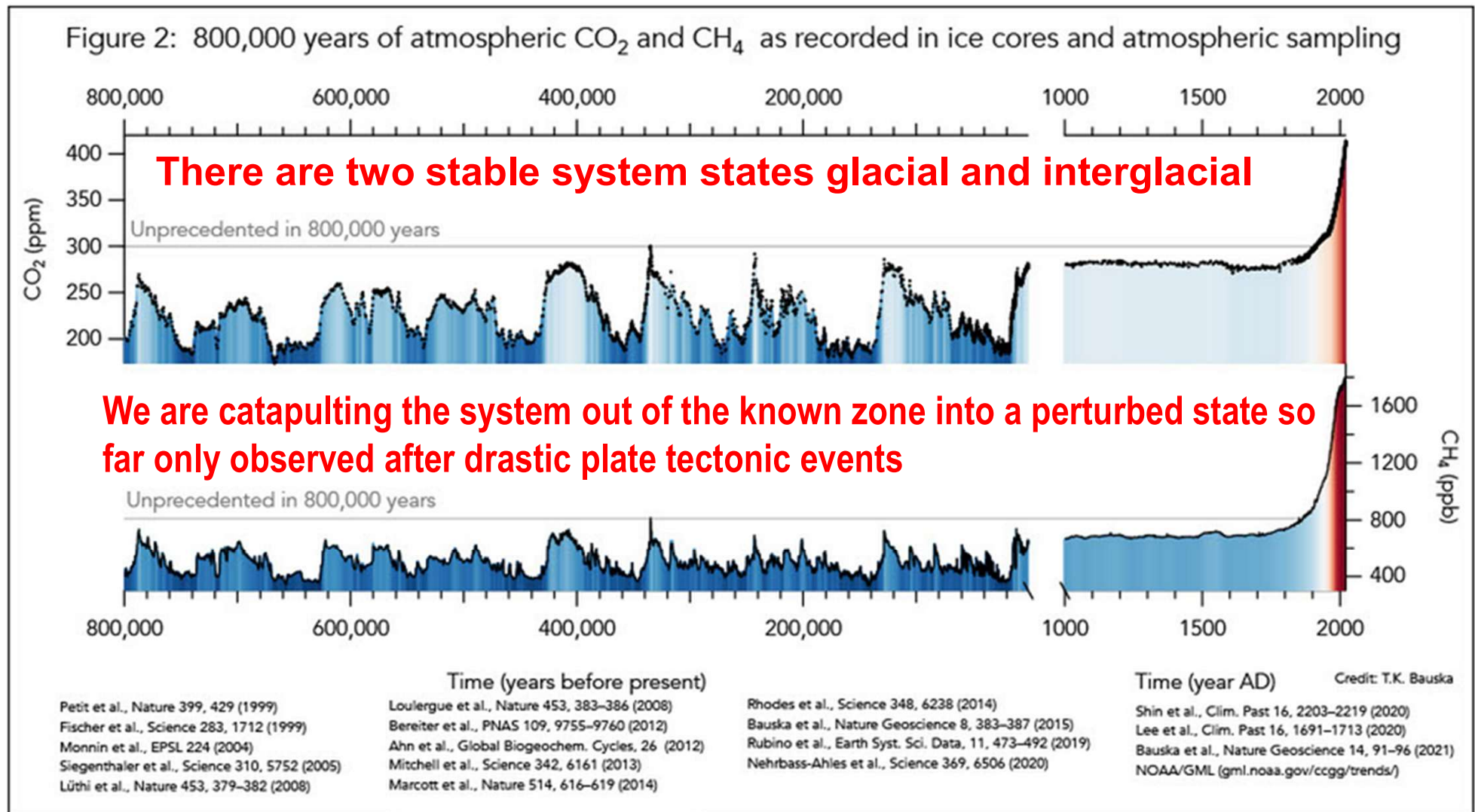


[npj Climate and Atmospheric Science](#) 8, Article number: 263 (2025) |

The gap periods are marked with gray-shaded areas. Numbers at the bottom indicate MIS. The Glacial Terminations are shown from TI to TIX.

**Preadthropogenic interglacial mixing ratios of  $N_2O$  were 270 to 280 ppbv**

# Antropogenic atmospheric composition is unprecedented throughout the Pleistocene



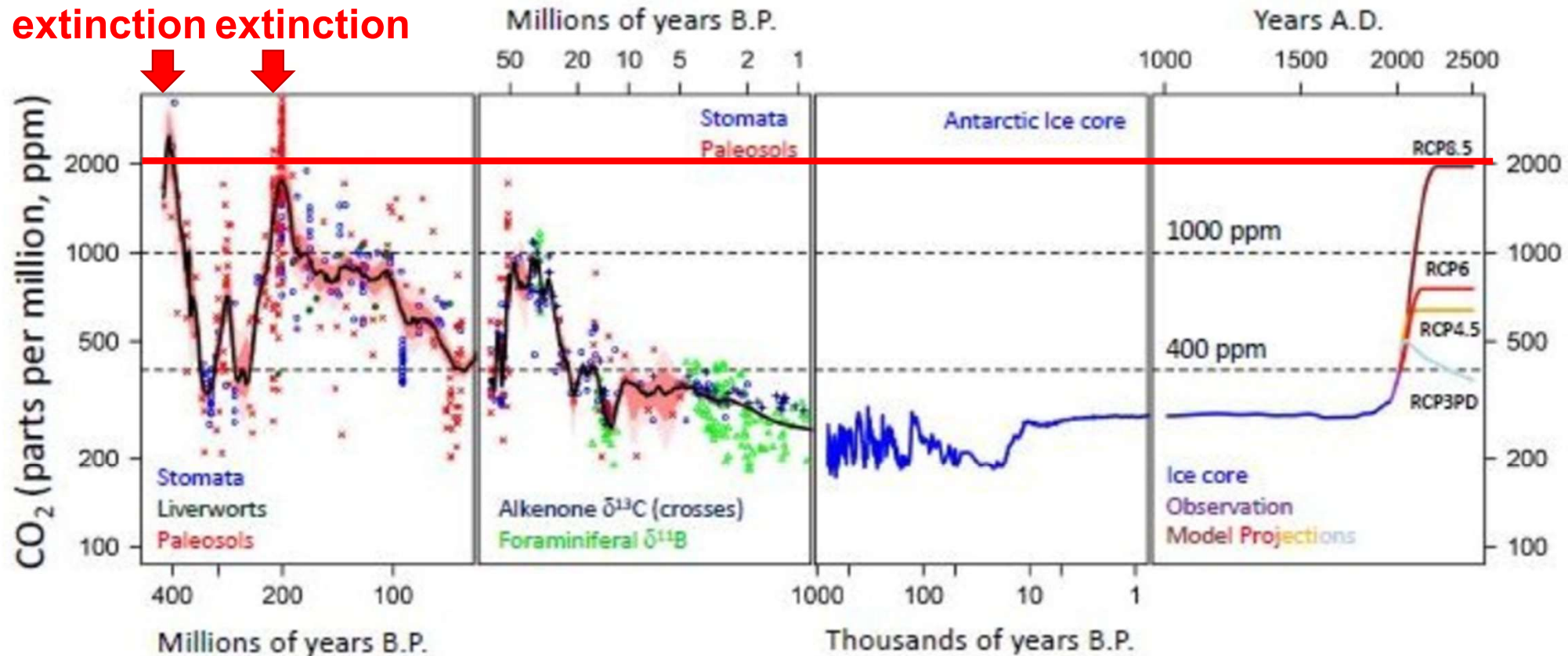
<https://www.bas.ac.uk/data/our-data/publication/ice-cores-and-climate-change/>

# The target of our current emission trajectory RCP8.5 is a major mass extinction

Both extinction were triggered by global warming, related to volcanism, and anoxia

O-S  
extinction

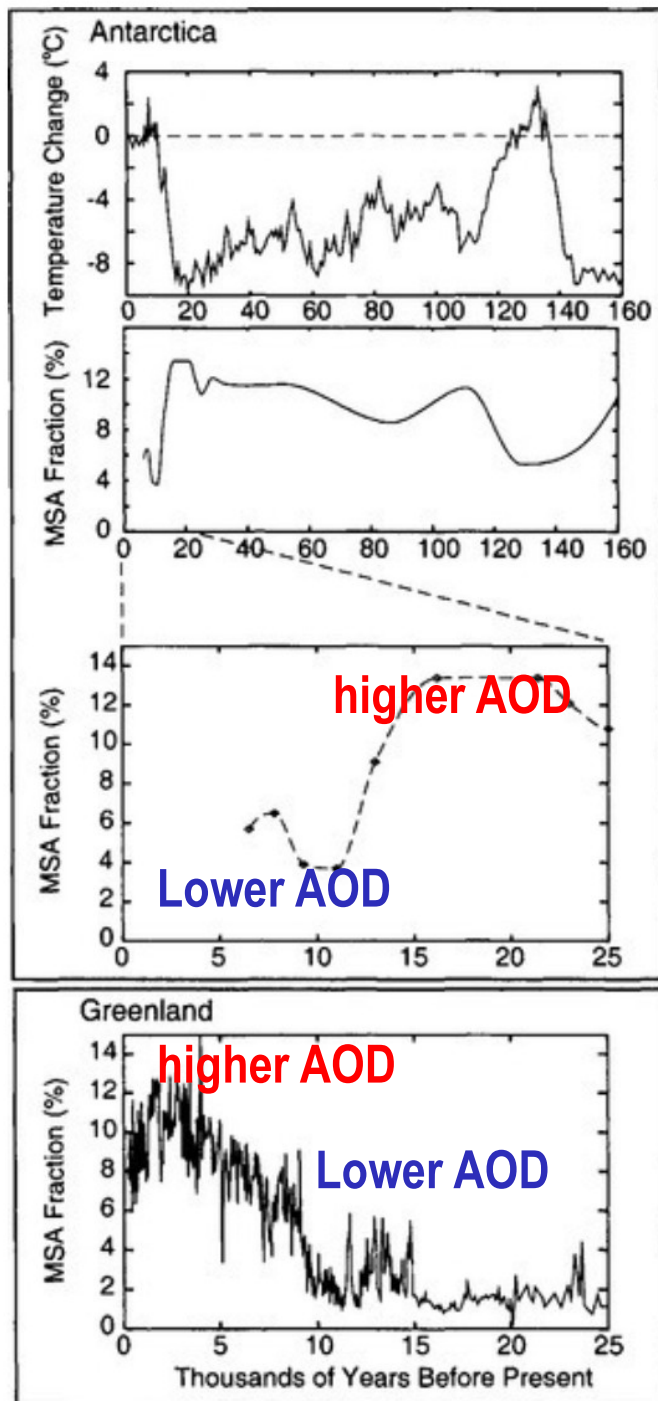
Tr-J  
extinction



Credit: Gavin Foster et al. (CC 3.0) [skepticalscience.com/Past-and-Future-CO2.html](http://skepticalscience.com/Past-and-Future-CO2.html)



# Ice age aerosol burden

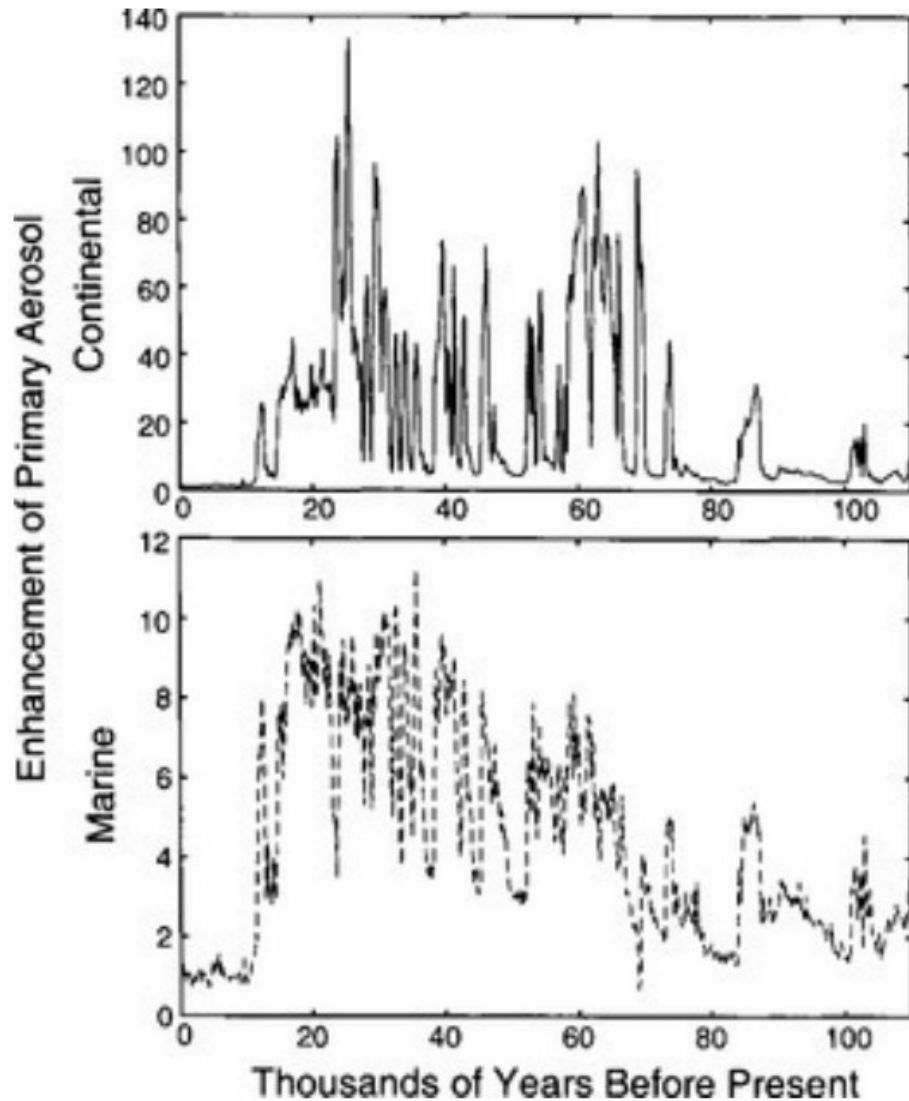


The North Atlantic appears to have higher biological productivity, DMS production and AOD during interglacials when the AMOC is relatively strong and lower biological productivity, DMS production and AOD during glacials when the AMOC is relatively weak.

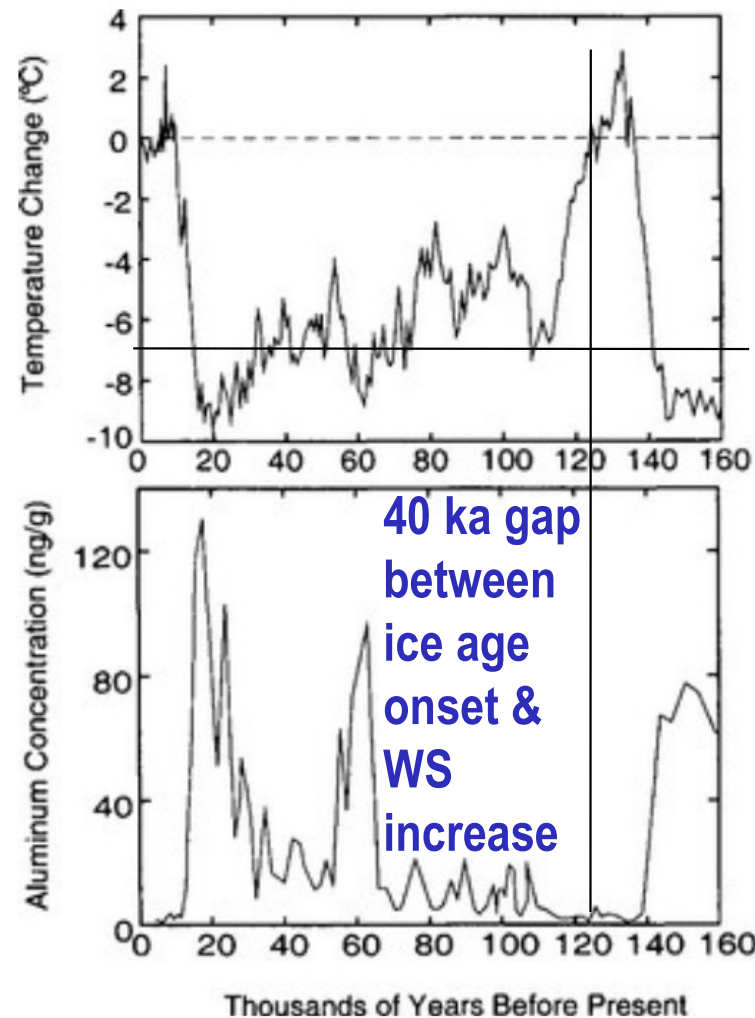
The Southern Ocean appears to have had higher biological productivity, DMS production and AOD during glacials when the dust transport and iron fertilization of the ocean were higher and lower biological productivity, DMS production and AOD during interglacials when dust transport is less. Productivity in the Southern ocean is more consistent with global CO<sub>2</sub> mixing ratio trends.

**FIG. 18-18** Top two panels show the history of Antarctic MSA fraction ([Legrand et al., 1991](#)), and coincident isotopic temperature history ([Jouzel et al., 1993](#)). Bottom two panels present a closer look at the last 25 kyr, for both Antarctica and Greenland ([Saltzman et al., 1997](#)).

# Ice age aerosol burden



**FIG. 18-19** Enhancement, relative to modern values, of marine and continental primary aerosols in Greenland through the last ice age (derived from [Mayewski et al., 1997](#)).



**FIG. 18-20** Correlation of temperature history ([Jouzel et al., 1993](#); [Salamatin et al., 1998](#)) and continent-derived dust reaching Antarctica ([De Angelis et al., 1987](#)).

The ice age atmosphere was more arid dustier in both hemispheres

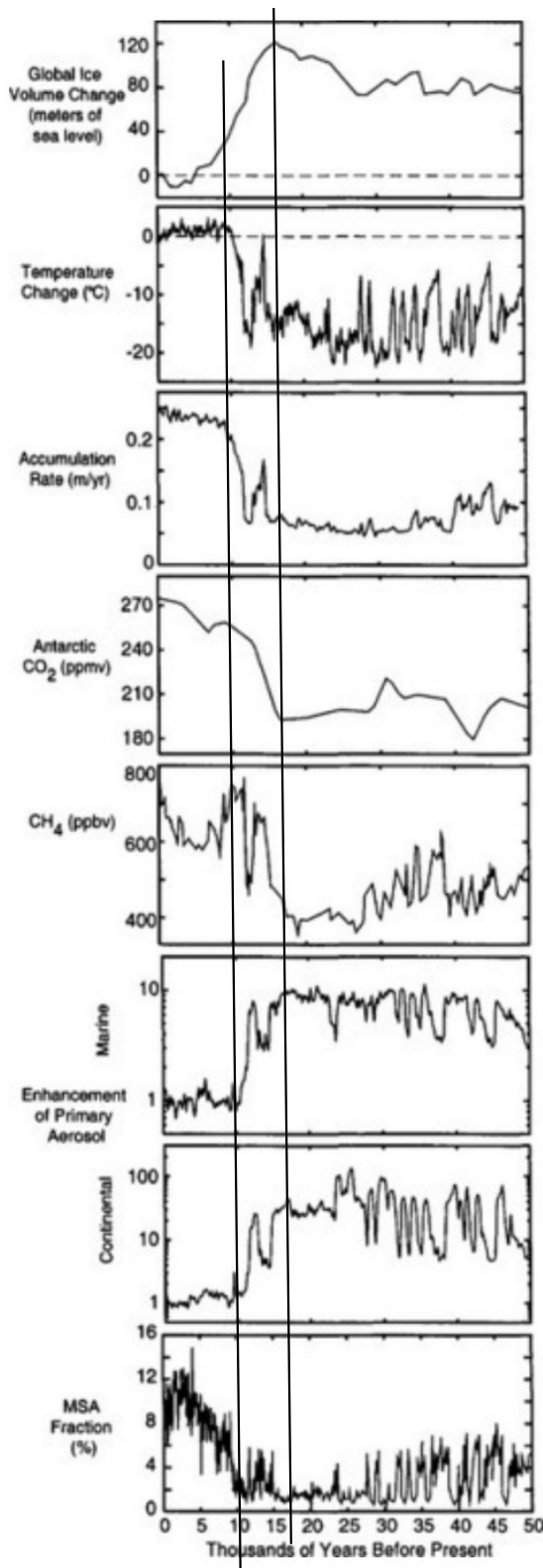
Wind speeds were higher during the glacial (sea salt deposition & dust grain size distribution are good proxies)

# The summary

- Noticeable ice melting and significant sea level rise occurs concurrent with increases atmospheric CO<sub>2</sub> and CH<sub>4</sub> mixing ratios but before local temperatures at the ice core drill sites and ice accumulation rates change.
- Maybe the initial melting is related to melting of lower latitude high altitude ice sheets.
- Accumulation rates, wind speeds & temperatures at the core drill site change rapidly but relatively late.
- The ocean productivity (MSA) in the North Atlantic which is related to AMOC strength picks up last.

**FIG. 18-22** The last 50 000 years of environmental history, recorded in central Greenland (GISP2 and GRIP ice cores), plus the CO<sub>2</sub> record from Vostok, Antarctica, and global ice volume measured as sea level depression. (From top to bottom, references are: [Shackleton, 1987](#) ; [Cuffey et al., 1995](#) and [Grootes et al., 1993](#) ; [Cuffey and Clow, 1997](#) ; [Chapellaz et al., 1990](#) ; [Brook et al., 1996](#) ; [Mayewski et al., 1997](#) ; [Saltzman et al., 1997](#) .)

- Some drastic temperature changes like the Younger Dryas event can happen in very short (decadal) time spans
- This event is likely related to a sudden strengthening of the AMOC triggering as strong Arctic warming followed by an AMOC collapse on account of increased melt water input.





# PRACTICE QUESTIONS

**You can submit the answer to these questions by Monday 15<sup>th</sup> September if you want to get them corrected before the Midsem exam**

# MCQs

1. We can use isotopes of oxygen to determine the climate of the past. Cool climates have higher ratios of  $\delta^{18}\text{O}$  in the shells of deep water organisms because:
  - a) The heavier  $^{18}\text{O}$  preferably evaporates from the ocean
  - b) The heavier  $^{18}\text{O}$  is preferably incorporated into the shell.
  - c) More isotopically light ice was trapped in the large continental ice sheets and the water in the oceans was heavier
  - d) The biomes changed and we study different organisms as ice age and warm age proxy
2. Which of the following is not a proxy used to reconstruct past climate?
  - a) Ship logs
  - b) Diatom fossils in sediment cores
  - c) Ice sheets
  - d) Thermometers
3. What causes the seasons on Earth?
  - a) The Earth's rotation
  - b) Weather changes
  - c) The tilt of the Earth's axis
  - d) The distance from the Moon
4. Why are ice age temperature records in Greenland ice cores much noisier than those in the Antarctic ice cores?
  - a) They were retrieved earlier the analytical accuracy is worse
  - b) Preservation conditions differ
  - c) The AMOC switched on and off several times throughout the glacial and this caused many rapid temperature fluctuations
  - d) The accumulation rates at the south pole are so slow that the temporal resolution of the core can not resolve Dansgaard-Oeschger events.

# MCQs

1. What are some examples of climate proxies?
  - a) Diatoms,  $\delta^{18}\text{O}$ , pollen
  - b) Ice cores, tree rings, sediment cores
  - c) Both A and B
  - d) None of the above
2. Which of the following is not a proxy used to reconstruct past climate?
  - a) Ship logs
  - b) Diatom fossils in sediment cores
  - c) Ice sheets
  - d) Thermometers
3. What causes the seasons on Earth?
  - a) The Earth's rotation
  - b) Weather changes
  - c) The tilt of the Earth's axis
  - d) The distance from the Moon
4. How have average temperatures have changed in different regions. The largest change in temperature occurred
  - a) At northern latitudes and over water
  - b) Around the equator over water
  - c) Around the equator over land



# Numerical (may need to revise lecture 10)

1. The ratio of  $^{18}\text{O}$  to  $^{16}\text{O}$  in the ocean water is about 2045:100000.
  - a) Suppose  $10^6$  atoms of  $^{16}\text{O}$  evaporate in equilibrium with ocean water at 298 K. How many  $^{18}\text{O}$  atoms will be associated with these  $^{16}\text{O}$  atoms in the atmosphere?
  - b) Suppose now that this air is shifted to higher latitudes and cooled to 293 K wherein half of the  $^{16}\text{O}$  atoms condense to form raindrops. What number of  $^{18}\text{O}$  atoms will condense and what number will remain in the atmosphere
  - c) Evaluate the  $\delta^{18}\text{O}$  for the precipitation that forms in (b) as well as the  $\delta^{18}\text{O}$  of the gas phase water left behind in the air parcel that moves further north.
  - d) Describe verbally what will happen to the rain that forms inside a continent as the air mass moves away from the ocean and the moisture rains out bit by bit while the air mass moves inland
  - e) What happens to the number of  $^{18}\text{O}$  atoms in the ocean if lot of water is removed from the ocean and deposited in ice sheets

# Numerical (may need to revise lecture 10)

1. A) At present the SST in the tropics is on average ~301 K while the polar ocean has a SST of ~277 K. Use:

$$\ln \left( \frac{P_{16}}{P_{18}} \right) = \frac{7.88}{T} - 0.0177$$

to calculate the  $^{18}\text{O}/^{16}\text{O}$  ratio of an atmosphere in equilibrium with the polar and tropical ocean, respectively. Use

$$\delta^{18}\text{O} = \left[ \frac{\left( \frac{^{18}\text{O}}{^{16}\text{O}} \right)_{\text{sample}}}{\left( \frac{^{18}\text{O}}{^{16}\text{O}} \right)_{\text{ref}}} - 1 \right] \times 1000$$

With the reference VSMOW=0‰ to calculate the  $\delta^{18}\text{O}$  composition of water vapor in equilibrium with the tropical and polar ocean in permille.

- B) Do you expect the average  $\delta^{18}\text{O}$  of water vapor during the ice age to be more negative or more positive.

# Long answer

1. Explain why methane emissions change between ice ages and warm ages and explain how these changes in the methane emission rate will impact the lifetime of methane in the atmosphere.
2. Explain how the ocean CO<sub>2</sub> sink changes between ice ages and warm ages. Which chemical, thermodynamic, physical and biological processes contribute towards altering ocean sink term.
3. Explain the difference between paleoclimate proxies and paleoclimate archives. List at least four paleoclimate proxies and two paleoclimate archives.
4. What triggered the ice ages and how did the climate change during the ice ages?
5. Why are gas bubbles trapped in ice younger than the surrounding ice? Which stable isotopes preserved in these gas bubbles do we use to detect rapid changes in climate and calibrate the age difference between the bubbles and the surrounding ice?
6. Why are the ice age temperature records in Greenland ice cores much noisier than those in the Antarctic ice core. What caused the more rapid temperature fluctuations in the North Atlantic.

# Long answer

1. Why does the ocean productivity in the Southern Ocean as recorded by MSA preserved in the ice cores respond differently to ice ages and warm ages compared to the ocean productivity in the North Atlantic?
2. Why are the quantity of dust aerosol and the quantity of marine aerosols trapped in the ice cores a proxy for ancient wind speeds?
3. What is wrong with the movie “Day after Tomorrow” in which an ice age starts rather suddenly (within a few days) due to sudden AMOC collapse. Based on the paleoclimate record: In which direction is the change sudden (and how sudden), in which direction is the change slow.